## 11.2.7 Wrap-Up: Growth Mindset

Making mistakes is one of the best ways to learn. Reflecting on our mistakes helps our brains to look for that mistake again and to avoid it the next time.

1. What mistake did you make today that you can learn from?

**Means**
**I**
**Start**
**To**
**Acquire**
**Knowledge**
**Experience**
**Skills**

# Unit 11, Lesson 3: Multiplying Rational Numbers

### Benchmark

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.

### Learning Target

☐ I can fluently multiply rational numbers.

## 11.3.1 Warm-Up: Bell Work

1. Using the digits 1 to 6 at most one time each, fill in the boxes so that the top two equations are equal, and the bottom equation has the greatest value.

$$^{-}\square + \square =$$

$$^{-}\square - \square =$$

$$^{-}\square - {}^{-}\square =$$

## 11.3.2 Collaboration: Signed Number Rules

1. Fill in the blank with the words *positive* or *negative* to describe the rules for multiplying integers.

   - The product of two positive integers is a ____________ integer.
   - The product of two negative integers is a ___________ integer.
   - The product of a negative integer and a positive integer is a ____________ integer.

2. Discuss with your partner whether you think the same rules apply to rational numbers. Write a summary of your reasoning.

3. Complete the table. Use a calculator to find the product of each expression to verify if the same rules for signs apply when multiplying rational numbers.

| Expression | Sign of Product Using the Rules | Product | Did the Same Rule Apply? |
|---|---|---|---|
| $\left(-\frac{3}{5}\right)\left(-\frac{3}{11}\right)$ | | | |
| $\frac{7}{18}\left({}^{-}4\frac{8}{25}\right)$ | | | |
| $21.285 \cdot {}^{-}3.8$ | | | |
| $({}^{-}91.09)({}^{-}6.4)$ | | | |

4. Complete the sentence.

The rules for multiplying rational numbers are

- ○ the same as
- ○ different from

the integer rules for multiplication.

## 11.3.3 Guided Practice: Applying Signed Number Rules to Rational Numbers

1. With your partner, decide who will be Partner A and who will be Partner B. Then, find the product of the expression in the column that matches your letter.

| Column A | Product | Column B |
|---|---|---|
| $790\left(\frac{1}{10}\right)$ | | $7.9(10)$ |
| $\left(-\frac{6}{7}\right) \cdot 7$ | | $(0.1) \cdot (-60)$ |
| $2.1 \cdot (-2)$ | | $(-8.4)\left(\frac{1}{2}\right)$ |
| $2.5 \cdot {}^{-}3.25$ | | $\left(-\frac{5}{2}\right)\left(\frac{13}{4}\right)$ |

The expressions in each row have the same product. Check your work with your partner and, if necessary, resolve any differences. Write the agreed-upon product in the center column.

2. Choose a value from Row A and a value from Row B that, when multiplied, will result in a negative answer. Then, calculate the product.

**Row A**

| | | |
|---|---|---|
| $-9\frac{2}{7}$ | $-\frac{3}{5}$ | $12\frac{2}{9}$ |

**Row B**

| | | |
|---|---|---|
| $4\frac{3}{8}$ | $\frac{11}{16}$ | $-\frac{9}{50}$ |

| **Row A** | | **Row B** | | **Product** |
|---|---|---|---|---|
| $\square$ | $\times$ | $\square$ | $=$ | $\square$ |

## 11.3.4 Wrap-Up: Ticket Out the Door

1. Find the product of each expression.

   a. $(0.24)({}^{-}3.8)$

   b. $\left(-\frac{2}{5}\right)\left(-\frac{9}{8}\right)$

2. Fill in the blanks with the words *positive* or *negative*.

   - Two positive factors result in a __________ product.
   - Two negative factors result in a __________ product.
   - A positive factor and a negative factor result in a ____________ product.